

# SHUVAM PAL

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## Education

**Dayananda Sagar College of Engineering**

**Bengaluru, India**

*B.E. Artificial Intelligence and Machine Learning*

## Experience

**Bentham AI**

*Dec 2025 – Feb 2026*

*Backend Engineering Intern: Go, Node.js, Docker, GCP, Puppeteer*

*Remote*

- Built a Puppeteer/Node.js automation service for MCA company registrations, handling retries, network-idle waits, API checks, and Google Vision CAPTCHA handling; reduced manual filing time by **85%**.
- Built a Go notification service with a single API for WhatsApp and email channels, MSG91 integration, request validation, and provider separation.
- Containerized backend services with Docker and deployed them to Google Cloud Run using repeatable deployment scripts.

**Google Summer of Code**

*May 2025 – Sept 2025*

*Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX*

*Remote*

- Built Python tooling to segment **95 brain subparts** from MRI images using PyTorch/ONNX model outputs and generated masks.
- Reduced processing time by **87%** using asynchronous functions and parallel execution across large MRI volumes.
- Improved 3D surface visualization and output inspection paths in the existing desktop medical-imaging application.

## Open Source

[Contributions](#)

- Contributed to Invesalious, IOOS, CRIU, and Tiled through maintainer-reviewed fixes, tests, and backend changes.
- Implemented Kuadrant MCP Gateway backend-outage grace handling with advertised-tool retention and recovery/removal test coverage.
- Updated OPA opa fmt behavior for multiline Rego arrays/objects with v0/v1 formatter golden tests.

## Projects

**Murdoc** | *Python, FastAPI, MCP, OpenTelemetry, Prometheus, Docker*

[GitHub](#)

- Built a FastAPI gateway for LLM, HTTP-tool, and MCP requests with route profiles, allow/block rules, decision logs, and audit records.
- Added RBAC settings, PII redaction, health checks, OpenTelemetry traces, and Prometheus metrics so backend behavior is easier to debug.

**Postificus** | *Go, Echo, PostgreSQL, Redis, RabbitMQ, Docker, Prometheus, Go-Rod*

[GitHub](#)

- Built a Go/Echo backend for content publishing workflows with REST APIs, PostgreSQL persistence, and account/platform state.
- Added RabbitMQ workers with retries and dead-letter queues for long-running publish jobs, plus Redis-backed draft and cache state.
- Added Docker setup, health checks, Prometheus metrics, and Go-Rod browser fallback for platforms without stable publish APIs.

## Technical Skills

**Languages:** Python, Go, TypeScript

**Framewor**ks: FastAPI, Echo, Node.js, Express

**Tools:** PostgreSQL, Redis, RabbitMQ, Docker, GitHub Actions, GCP, Prometheus, OpenTelemetry, Go-Rod, Puppeteer